

Luis Lauro Aizpuru-Vargas, M.Sc.

✉ A01195658@tec.mx  Luis Lauro Aizpuru-Vargas
 <https://luisaizpuru-vargas.github.io/>

Employment History

- 2019 – 2020  **Research Assistant**
Jülich Supercomputing Centre (JSC)
Forschungszentrum Jülich
Jülich, Germany
- 2019  **Teaching Assistant**
Argelander-Institut für Astronomie (AIfA)
Universität Bonn
Bonn, Germany

Education

- 2024 – 2028  Ph.D. Computer Science (Medical Informatics Track)
Tecnológico de Monterrey
Monterrey, Mexico
Preliminary Thesis title: *“Personalized AI-Driven Decision Support for Sedation and Analgesia in the ICU in the Alison++ Project”*
- 2021 – 2023  M.Sc. Medical Physics
Pontificia Universidad Católica de Chile
Santiago, Chile
Thesis title: *“A study of the vascular effect of photodynamic therapy with oxygen variation on chorioallantoic membrane model”*
- 2017 – 2020  M.Sc. Astrophysics
Rheinische Friedrich-Wilhelms-Universität Bonn
Bonn, Germany
Thesis title: *“Investigation on gravitational star-disk encounters producing ‘Oumuamua and Borisov-like Interstellar Objects”*.
- 2012 – 2017  B.Sc. Physics (Astrophysics Track)
University of Alberta
Edmonton, Canada

Research Publications

Journal Articles

-  **L. Aizpuru-Vargas**, R. Villalba-Pacheco, R. Álvarez-Vaccaro, S. Huichacura, J. T. Egaña, B. Sánchez-Nieto, I. Espinoza, and H. Harb Buzzá. “Photosynthetic microalgae-enhanced oxygenation amplifies vascular response to photodynamic therapy in a chorioallantoic membrane model.” *Photochemical & Photobiological Sciences*. Vol. 25. 2026

- 2 S. Pfalzner, L. Aizpuru-Vargas, A. Bhandare, and D. Veras. “Significant interstellar object production by close stellar flybys.” *Astronomy & Astrophysics*. Vol. 651. 2021

Skills

Languages	Spanish (Native Fluency) English (Very High Fluency) Portuguese (High Fluency) German (Moderate Fluency) French (Basic Fluency)
Coding	Python, R, SQL, $\text{\LaTeX}$
Technical Skills	Advanced Mathematical Proficiency Scientific Modeling Data Analysis
Transferable Skills	Problem-Solving Research and Critical Thinking Attention to Detail Communication Multidisciplinary Interests

Miscellaneous Experience

Awards and Achievements

2024-2028	PhD Stipend SECIHTI (Secretariat of Science, Humanities, Technology and Innovation) Mexico
2021-2023	Graduate Studies Academic Scholarship Pontificia Universidad Católica de Chile
2019-2020	Research Stipend Forschungszentrum Jülich
2012-2015	Undergraduate Studies Academic Scholarship University of Alberta

Certifications

2023-2024	Data Science Bootcamp Awarded by Universidad del Valle de México. Clinical SBRT and Radiosurgery Awarded by Elekta Foundation and RCC.
2023	Advanced Radiotherapy and Protontherapy Awarded by NHS England. Certificate in Radiological Security Awarded by COFEPRIS.
2022	Scientific Workshop on Emerging Techniques in Radiotherapy Awarded by DKFZ.
2021	Workshop on Computational Methods in Radiotherapy Awarded by DKFZ.

References

Referee 1 Prof. Anna Karen Gárate-Escamilla School of Engineering and Sciences Tecnológico de Monterrey karen.garate@tec.mx	Referee 2 Prof. Hilde Harb-Buzzá Institute of Physics PUC Chile hilde.buzza@uc.cl	Referee 3 Prof. Susanne Pfalzner Jülich Supercomputing Centre Forschungszentrum Jülich s.pfalzner@fz-juelich.de
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